PHYSICS-CONSISTENT GENERATIVE INVERSE DESIGN OF ANGLE-RESOLVED FOURIER-OPERATOR METASURFACES

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Abstract. We present a maintainable manuscript draft centered on physics-consistent generative inverse design of angle-resolved Fourier-operator metasurfaces. The governing thesis is that freeform Fourier-operator metasurface design should be formulated in a unified angle–frequency–polarization response space rather than as a collection of disconnected task-specific optimizations. Within this formulation, the inverse problem becomes conditional generation of physically admissible structures from target response fields, with physical consistency enforced through simulation-grounded supervision, candidate verification, and post-generation reranking. The resulting narrative emphasizes a unified design paradigm: second-order differentiation, polarization-selective operators, and spectrally structured transfer functions are treated as different slices of the same response-space objective rather than as separate application libraries.

1. Introduction

1.1. From Fourier optics to angle-resolved operator metasurfaces

Fourier optics provides the formal language for interpreting imaging systems and linear spatial operators in the frequency domain, where filtering and differentiation are encoded through transfer functions rather than through pointwise geometric manipulations [4]. In metasurface-enabled analog photonics, this viewpoint has become especially powerful because a subwavelength structured interface can be designed to emulate a desired optical operator within an extremely compact footprint. Recent demonstrations of mathematical metasurfaces and nonlocal operator devices have shown that differentiation, edge enhancement, and polarization-dependent image processing can be implemented by engineering the angular response of nanostructured optical layers [13, 1, 7, 3].

For angle-resolved Fourier-operator metasurfaces, however, the relevant design object is not merely a scalar transmission efficiency or a single transfer-function slice at normal incidence. The device must realize a structured optical response that varies jointly with incidence angle, wavelength, and polarization. This coupled dependence is not a secondary implementation detail; it is the mechanism by which operator fidelity, angular bandwidth, and polarization selectivity are actually realized. Accordingly, a satisfactory inverse-design framework should be able to represent and manipulate the response field itself.

1.2. Why inverse design remains difficult

Despite rapid progress in metasurface inverse design, freeform Fourier-operator devices remain challenging because the structure-to-response map is high-dimensional, strongly nonlinear, and frequently non-unique. A large number of distinct binary patterns can yield similar

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coarse operator behavior while differing substantially in spectral robustness, angular stability, or polarization leakage. Brute-force scanning is therefore prohibitively inefficient, and purely deterministic regression tends to average over valid design modes instead of preserving solution diversity [12, 8, 14].

The challenge becomes more pronounced when the target is defined over a multi-axis response field. Conventional design workflows often reduce the problem to one operating wavelength, one incidence condition, or one target task at a time. Such simplifications are useful for isolated demonstrations, but they obscure the central question addressed in this work: whether angle-resolved Fourier-operator metasurface design can be cast as a unified response-space inverse problem whose different practical instantiations emerge from one common formulation.

1.3. Central thesis and scope of this paper

The central thesis of this manuscript is that physics-consistent generative inverse design offers a natural framework for this unification. We treat the desired metasurface behavior as a target optical response field $\mathbf{R}^{\star}(\lambda, \theta, p)$ defined over wavelength, incidence angle, and polarization channel. The inverse problem is then to generate one or more physically plausible structures whose verified response matches this target field under full-wave or RCWA-based evaluation. Under this formulation, second-order differentiation, polarization-multiplexed image processing, and spectrally selective operator behavior are not separate "tasks" but different constraints on the same unified response object.

This narrative deliberately avoids framing the work as a platform paper built around a growing library of independent demonstrations. Instead, the paper is organized around a single design paradigm: angle–frequency–polarization response-space engineering enforced through a physics-consistent generative loop. The sections that follow formalize this viewpoint, describe the generative framework, and use representative results to argue that unification is the main scientific contribution.

2. Unified Angle–Frequency–Polarization Response Formulation

2.1. Response field as the primary design object

Let $\mathcal{G}$ denote a freeform binary metasurface pattern defined on a fabrication-compatible lattice. Instead of summarizing its behavior by a single figure of merit, we associate to each structure a response tensor

$$\mathbf{R}(\mathcal{G}) = \{T_{pp}(\lambda_i, \theta_j), T_{ss}(\lambda_i, \theta_j)\}_{i=1,\ldots,N_\lambda,\, j=1,\ldots,N_\theta},$$

where the two channels can be generalized to a broader polarization basis when needed. This object records the operator-relevant transfer characteristics over a finite angle–frequency sampling window and therefore serves as the physically meaningful state variable for inverse design.

This response-centered definition is particularly appropriate for Fourier-operator metasurfaces. In such devices, the desired functionality is encoded by the shape of a transfer function across spatial-frequency coordinates, which are directly coupled to angle under the illumination and imaging geometry. Once the design objective is written in this way, the traditional distinction between "differentiator," "filter," and "polarization-selective operator" becomes a distinction between target field geometries rather than between fundamentally different inverse problems.

2.2. Unified target families in a single formulation

Within the proposed response-space viewpoint, several seemingly different device categories can be represented without changing the mathematical formulation. A second-order differentiator is expressed through a prescribed even-symmetric angular transfer profile, potentially over multiple wavelengths. A polarization-multiplexed operator is obtained by assigning different target profiles to different polarization channels. A spectrally selective operator corresponds to a structured dependence of the same profile family on λ. In all cases, the optimization variable remains $\mathcal{G}$ and the target remains $\mathbf{R}^\star(\lambda, \theta, p)$.

This observation is not merely organizational. It clarifies that the role of a unified generative model is to learn a conditional distribution over structures given target response fields that may vary in shape, anisotropy, and polarization dependence, but still belong to one common response manifold. That is the sense in which this paper argues for a unified angle–frequency–polarization design paradigm rather than a catalog of unrelated demonstrations.

2.3. Physics-consistent inverse-design objective

The inverse-design objective is to identify structures whose simulated response $\mathbf{R}_{\mathrm{sim}}(\mathcal{G})$ matches the target field while respecting physical and fabrication constraints. We therefore write the score function as

$$\mathcal{S}(\mathcal{G}; \mathbf{R}^\star) = \alpha\, \mathcal{L}_{\mathrm{resp}}(\mathbf{R}_{\mathrm{sim}}, \mathbf{R}^\star) + \beta\, \mathcal{L}_{\mathrm{pol}} + \gamma\, \mathcal{L}_{\mathrm{spec}} + \delta\, \mathcal{L}_{\mathrm{phys}},$$

where the leading term measures response-field mismatch and the additional terms encode polarization separation, spectral smoothness or selectivity, and physical admissibility. The precise weighting may vary across experiments, but the formal structure remains unchanged because all objectives are defined in the same response space.

Equally important, the formulation is compatible with multi-solution inversion. Because many structures can satisfy the same operator-level constraint within a tolerance band, the objective should not be interpreted as seeking a unique geometry. Rather, it defines a target-conditioned feasible set. This motivates the generative perspective adopted in the next section: instead of directly regressing a single structure, we model the conditional distribution of valid solutions and impose physics consistency through simulation-grounded evaluation.

3. Physics-Consistent Generative Framework

3.1. Simulation-grounded dataset construction

The framework begins with a dataset $\mathcal{D} = \{(\mathcal{G}_k, \mathbf{R}_k)\}_{k=1}^N$ built from freeform binary patterns and their corresponding electromagnetic responses generated by RCWA or full-wave solvers [10, 11]. The key design choice is that the supervision target is not a compressed scalar metric but the response field itself. This preserves the angular, spectral, and polarization structure that defines operator behavior and prevents the learning problem from collapsing into a task-specific surrogate objective.

In practice, dataset diversity matters at two levels. Geometric diversity is needed to expose the generator to multiple design modes for similar responses, while response diversity is required to populate the angle–frequency–polarization manifold with enough variety to support conditional generation. This dual diversity is central to physics-consistent generative inverse design because the model must capture both the multimodality of the inverse map and the physical regularity of the forward map.

3.2. Forward surrogate and candidate concentration

To reduce direct electromagnetic evaluation cost, we introduce a forward surrogate $f_\phi : \mathcal{G} \mapsto \widehat{\mathbf{R}}$ trained to approximate the response field associated with each candidate structure. The role of this model is not to replace physical verification but to improve candidate concentration by filtering out clearly implausible designs and prioritizing samples whose predicted response lies near the target manifold. This surrogate-assisted stage effectively reshapes the sampling distribution seen by the downstream generative loop.

Such a forward model is especially valuable for angle-resolved operator metasurfaces because evaluation over a dense angle–frequency grid can be substantially more expensive than scalar performance estimation. By predicting the full response field, the surrogate can be used for proposal screening, difficulty-aware curriculum construction, and approximate ranking, while final acceptance remains anchored to the electromagnetic solver.

3.3. Conditional generative inversion

The inverse stage is modeled as a conditional distribution $p_\theta(\mathcal{G} \mid \mathbf{R}^\star)$, instantiated here as a diffusion-based generator motivated by recent progress in metasurface generative design [15, 16, 5]. Diffusion is attractive in this context because it naturally supports one-to-many conditional synthesis and can preserve structural diversity without forcing all feasible solutions toward a single averaged design. This is particularly important when multiple binary patterns realize comparable operator fields but differ in fabrication margin or off-target robustness.

Conditioning is applied at the level of the target response field rather than at the level of a discrete task label. As a result, the generator is explicitly asked to infer structures consistent with a continuous optical specification over angle, frequency, and polarization. This design choice is aligned with the scientific framing of the paper: the model is not choosing among pre-defined applications, but synthesizing structures from a unified response-space description.

3.4. Closed-loop physical verification and reranking

Physics consistency is enforced through a closed loop in which generated candidates are re-evaluated by the electromagnetic solver and reranked according to the verified score $\mathcal{S}(\mathcal{G}; \mathbf{R}^\star)$. This final stage is essential because visually plausible or surrogate-consistent patterns can still violate the true Maxwell response. The reranking step therefore acts as a correction mechanism that restores physical authority to the solver while retaining the efficiency benefits of generative sampling.

The practical consequence is that the framework produces not just a nominal inverse solution but a structured candidate set whose members can be compared in the same response space used to define the target. In this sense, physical verification is not an afterthought but a constitutive part of the generative design loop. The framework is therefore best understood as physics-consistent generative inverse design rather than as a pure data-driven generator followed by optional simulation.

4. Experimental Setup

4.1. Device family and parameterization

We consider freeform binary Fourier-operator metasurfaces parameterized on a fixed-resolution planar grid compatible with the intended fabrication process. The geometric representation is intentionally general enough to support nontrivial operator responses without biasing the search toward a narrow analytic motif. This choice aligns with the claim of unified design: if the formulation is truly response-driven, it should not rely on manual geometry specialization for each target family.

All targets are defined over a prescribed angle–frequency window and, when relevant, over multiple polarization channels. The primary benchmark remains the second-order Fourier differentiator because it provides a stringent but interpretable test case. Additional validations are then introduced not as independent application cases, but as controlled perturbations of the same unified response-space formulation.

4.2. Target specification and sampling protocol

The target field $\mathbf{R}^{\star}$ is discretized on a finite set of wavelengths and incidence angles selected to balance operator fidelity with simulation cost. The discretization is dense enough to encode the intended transfer-function geometry and to capture spectral or polarization-dependent deviations that would be invisible in single-slice optimization. This sampling strategy ensures that the learning and evaluation pipelines are tied to the same underlying response-space object.

For each target family, we generate multiple candidate structures per condition and carry them through surrogate screening and physical reranking. The goal is not only to identify the top design, but also to quantify how reliably the generator populates the target-conditioned feasible region. This candidate-level analysis becomes particularly informative when comparing deterministic and generative baselines.

4.3. Evaluation metrics and baseline models

We report three classes of metrics. The first class evaluates response-level agreement between the target and the verified field, including normalized response mismatch, transfer-function similarity, and channel-isolation error when polarization multiplexing is involved. The second class evaluates physically verified device performance, such as robustness across the sampled angle band and spectral stability. The third class evaluates operator-level behavior in downstream image-processing tests, where appropriate.

Baseline models include deterministic inverse regression, conditional variational autoencoders, conditional adversarial generators, and ablated versions of the proposed pipeline without physical reranking or without full response-field conditioning [8, 14, 9, 15]. This selection is intended to isolate the contribution of physics-consistent generative inversion rather than to expand the paper into an application benchmark suite.

5. Main Results

5.1. Second-order Fourier-operator metasurface as the anchor demonstration

We first examine the primary benchmark: a freeform metasurface designed to realize a second-order Fourier-operator response over a prescribed angular window. This example is used as the anchor demonstration because it requires the transfer function to satisfy a structured, physically meaningful profile rather than a single scalar criterion. The proposed framework produces candidates whose verified response closely tracks the target field and whose operator-level image transformations reproduce the expected second-derivative behavior.

The most important observation is not simply that a high-performing differentiator can be obtained. Rather, the verified response maps indicate that the generated structure captures the full angular operator geometry with much greater fidelity than models trained only against reduced objectives. This supports the central premise that response-field conditioning and physical reranking are essential when the design objective is intrinsically angle resolved.

5.2. Response-level verification

To evaluate the main design, we compare the target field, surrogate prediction, and physically verified response on the same angle–frequency grid. The surrogate is useful for concentrating candidates, but the physically verified map reveals the true quality of the inverse solution. In the strongest candidates, the verified response preserves both the target curvature and the intended even-symmetry structure across the operating band, indicating that the generator has learned a nontrivial correspondence between target operator shape and freeform geometry.

We also observe that candidates with similar scalar matching scores may differ substantially in their off-axis behavior. This is precisely why response-field evaluation is more informative than isolated operating-point metrics. By retaining the full target geometry in the evaluation protocol, the framework distinguishes designs that are merely locally adequate from designs that faithfully realize the intended operator over the specified response domain.

5.3. Operator-level behavior and physical plausibility

Image-domain tests provide an intuitive but secondary validation of the design. When the verified metasurface response is applied to representative inputs, the output reproduces the expected edge-enhanced or second-derivative-like patterns, confirming that the response-level agreement translates into operator-level function. More importantly, the physical reranking stage suppresses candidates that appear structurally plausible yet exhibit distorted transfer behavior under verification.

Taken together, these results support the claim that physics-consistent generative inverse design is not only able to synthesize candidate geometries, but able to do so in a way that preserves the intended operator physics. The anchor demonstration therefore serves as evidence for the larger response-space paradigm rather than as an isolated showcase.

6. Unified Validation Across Angle, Frequency, and Polarization

6.1. Angle-domain validation

The first axis of validation concerns angular response. We test whether the framework can preserve target transfer-function structure over finite incidence-angle windows rather than only at a single nominal point. The resulting verified fields show that the same response-conditioned generator used for the main differentiator can accommodate alternative angular envelopes and operator widths without changes to the underlying inverse-design formulation.

This observation is important because angle-domain control is the bridge between metasurface scattering physics and Fourier-operator functionality. A design framework that cannot generalize across angular target shapes would remain tied to isolated demonstrations. The present results instead suggest that angle dependence is naturally handled when the target is represented directly as part of the response field.

6.2. Frequency-domain validation

We next consider frequency-structured targets in which the desired operator behavior varies systematically across wavelength. Rather than treating each wavelength as a separate inverse-design problem, the generator is conditioned on a single response field whose spectral slices jointly define the objective. The resulting candidates reveal that spectrally selective operator behavior can be incorporated without changing the overall framework or introducing ad hoc task-specific branches.

This frequency-domain result reinforces the main conceptual claim of the paper. Spectral selectivity is not an “additional task” layered onto a baseline pipeline; it is another projection of the same angle–frequency–polarization design space. The framework succeeds to the extent that it preserves this coupling and uses it to synthesize physically admissible structures.

6.3. Polarization-domain validation

Finally, we validate the framework on polarization-dependent targets in which distinct transfer-function profiles are assigned to different polarization channels. Previous works on angle-multiplexed and dual-polarization metasurfaces have demonstrated the physical feasibility of such behavior [6, 7, 2]. Here the important point is that polarization selectivity enters through the same response-space representation already used for the main differentiator.

The verified results indicate that polarization channel separation can be learned and enforced within the unified framework while preserving operator fidelity. This cross-axis consistency is central to the paper's scientific positioning. The design pipeline is therefore not a library of angle, spectral, and polarization modules, but a single physics-consistent inverse-design method operating on a unified response manifold.

7. Ablation and Analysis

7.1. Effect of physical reranking

The first ablation removes post-generation physical reranking and selects candidates directly from the generative or surrogate stage. This degradation consistently increases the gap between nominal and verified performance, even when the chosen designs appear visually regular. The result confirms that physical reranking is not merely a refinement heuristic; it is a necessary component for aligning the generative model with the true electromagnetic response.

This finding also clarifies why the paper uses the term *physics-consistent*. The generative model alone does not guarantee consistency, particularly in the presence of solution multimodality and sparse sampling of the response manifold. Closed-loop verification is therefore required to transform a generative proposal mechanism into a reliable inverse-design system.

7.2. Effect of unified response conditioning

In a second ablation, we replace the full angle–frequency–polarization target with reduced conditioning signals, such as single-slice objectives or compressed scalar descriptors. These variants typically recover part of the desired operator behavior but fail to preserve the coupled structure of the verified response. Angular distortions, spectral drift, or polarization leakage become more pronounced when the target is oversimplified.

This behavior directly supports the response-space thesis of the manuscript. If the objective is compressed too aggressively, the inverse model no longer receives the information required to disambiguate physically distinct solutions. Full response conditioning therefore contributes not just numerical accuracy, but conceptual coherence between the target specification and the underlying device physics.

7.3. Generative versus deterministic inverse mapping

A final comparison examines deterministic regressors against conditional generative models. Deterministic baselines often produce reasonable average responses, but they exhibit reduced diversity and weaker best-case performance after verification. By contrast, the generative model provides a candidate ensemble whose members explore multiple regions of the feasible design set, allowing physical reranking to select geometries with improved verified fidelity.

This distinction is especially important for freeform Fourier-operator metasurfaces, where non-uniqueness is a feature of the problem rather than an implementation nuisance. The benefit of generative inversion is therefore not only that it samples more candidates, but that it respects the one-to-many character of the inverse map defined in the unified response space.

8. Discussion

8.1. Why unification matters

The main outcome of this study is conceptual as much as algorithmic. By placing angle, frequency, and polarization on equal footing within the target response field, the paper reframes Fourier-operator metasurface design as a single inverse problem with multiple physically meaningful projections. This shift matters because it moves the discussion away from isolated demonstrations and toward a more transferable understanding of what should be learned, generated, and verified in freeform optical operator design.

Under this interpretation, physics-consistent generative inverse design is not a generic machine-learning wrapper applied to a metasurface problem. It is a modeling choice tailored to the structure of the inverse map: the solution set is multimodal, the objective is field-valued, and physical verification remains indispensable. The combination of these ingredients is what makes the framework scientifically coherent.

8.2. Limitations and open questions

Several limitations remain. First, the present formulation emphasizes response-field matching and does not yet exhaust the richer complex-field information that may be required for some operator families. Second, discretization over angle and wavelength inevitably constrains what is learned and verified, particularly for devices with sharp resonant structure. Third, while the freeform binary parameterization is useful for demonstrating unified inversion, additional fabrication-aware constraints may be needed for transfer to a specific process line.

These limitations are best understood as opportunities for extending the same response-space philosophy rather than reasons to abandon it. Future work can incorporate phase-sensitive objectives, uncertainty-aware target conditioning, and fabrication-tolerant generative priors while preserving the central structure of the framework introduced here.

9. Conclusion

We have reorganized this manuscript around a single scientific claim: angle-resolved Fourier-operator metasurface design can be formulated as a unified angle–frequency–polarization response-space inverse problem and addressed through physics-consistent generative inversion. In this framework, the important object is the target response field, not a task label or a single operating-point metric. Freeform structures are generated conditionally from that field, screened by forward prediction, and ultimately selected through physical verification and reranking.

This perspective positions the work as a unified design paradigm rather than as a platform paper assembled from many unrelated demonstrations. The representative results across angle, frequency, and polarization validate that the same formulation can support multiple operator behaviors without changing its conceptual core. More broadly, the manuscript suggests that future progress in freeform metasurface inverse design may depend less on enumerating additional tasks and more on learning physically structured response manifolds with faithful closed-loop verification.

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